

Atomic-Scale Insights Into the Thermal Stability of High-Entropy Nanoalloys

Syrine Krouna, Anissa Acheche, Guillaume Wang, Nathaly Ortiz Pena, Riccardo Gatti, Christian Ricolleau, Hakim Amara,* Jaysen Nelayah,* and Damien Alloyeau*

High entropy alloy nanoparticles bring hope to developing more efficient nanomaterials for high-temperature applications. Nevertheless, the enhanced thermal stability of nearly equiatomic nanoalloys containing at least 5 metals is nothing more than theoretical speculation about the impact of thermodynamic contributions on their structural properties and remains to be proven. Here, in situ aberration-corrected scanning transmission electron microscopy (STEM) and molecular dynamics simulations are combined to investigate at the atomic scale the thermal behavior of AuCoCuNiPt nanoparticles (NPs) from 298 to 973 K. Both in situ STEM heating and atomistic simulations reveal strong structural and chemical evolutions in the NPs with the formation and melting of an AuCu layer at the surface of NPs at high temperature. This phase separation that appears progressively with temperature is driven by pronounced atomic diffusion that is surprisingly more active in these quinary nanoalloys than in monometallic and bimetallic subsystems. Besides ruling out the existence of sluggish diffusion in AuCoCuNiPt nanoalloys and lowering their temperature range of application, the study allows distinguishing kinetic and thermodynamic effects on their structural properties, which is an essential prerequisite to better control the synthesis of complex nanomaterials.

multi-element nanoparticles (NPs), also known as high-entropy nanoalloys (HENA), which consist of nanostructures composed of five or more metals, in equal or near-equal atomic ratios, crystallized in a solid solution.^[1] Although many physical and chemical fabrication methods have been developed^[2] there are still many gray areas concerning the synthesis-structure-properties relationships in these complex nanomaterials. The synergy of several effects makes HENA potentially efficient and stable nano-catalysts: the “cocktail” effect due to the interaction between the different elements can give rise to enhanced reactivity which counterbalances the low content of noble metals in the nanostructure. The variety of elements and atomic sizes generate many lattice distortions and crystalline defects that can create preferential surface reaction sites.^[3] Furthermore, the alleged sluggish atomic diffusion in these nanocrystals, together with the high entropy effects that slow down metal segregation, is expected to reinforce their

structural and chemical stability.^[4] Consequently, HENA are considered as promising nanostructures for a wide diversity of catalytic applications. Their presumably high structural stability makes HENA particularly interesting for high-temperature reactions, between 773 and 1073 K, such as the ammonia decomposition reaction to generate hydrogen^[5] and the ammonia oxidation reaction, to ensure a high nitrogen oxide selectivity.^[2f] Also, in order to reduce carbon monoxide (CO) emissions of car exhausts, HENA make propitious catalysts for the CO oxidation reaction^[6] in automotive engines that have operational temperature of 1173 K. Besides, their physical properties make HENA convenient thermoelectric materials for waste heat recovery or sources of either medium (473–923 K) or high (923–1923 K) temperature ranges.^[1c,7]

In this context, studying the thermal behavior of HENA is a necessary first step to evaluate their structural stability with the view to better target their potential high-temperature applications. Moreover, our current state of knowledge on atomic diffusion in HENA is still far from complete, and experimental evidence of the existence of sluggish diffusion in HENA is lacking. In the present work, we used in situ scanning transmission electron microscopy (STEM), corroborated with atomistic simulations, to study in real-time and at the atomic scale the structural

1. Introduction

To extend the potential of nanoalloys in different fields such as catalysis or mechanics, much attention has recently been paid to

S. Krouna, A. Acheche, G. Wang, N. O. Pena, C. Ricolleau, H. Amara, J. Nelayah, D. Alloyeau
Laboratoire Matériaux et Phénomènes Quantiques
Université Paris Cité – CNRS
Paris 75013, France
E-mail: hakim.amara@onera.fr; jaysen.nelayah@u-paris.fr; damien.alloyeau@u-paris.fr

A. Acheche, R. Gatti, H. Amara
Laboratoire d'Etudes des Microstructures
UMR 104 CNRS-ONERA
Université Paris-Saclay
Châtillon 92320, France

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adma.202414510>

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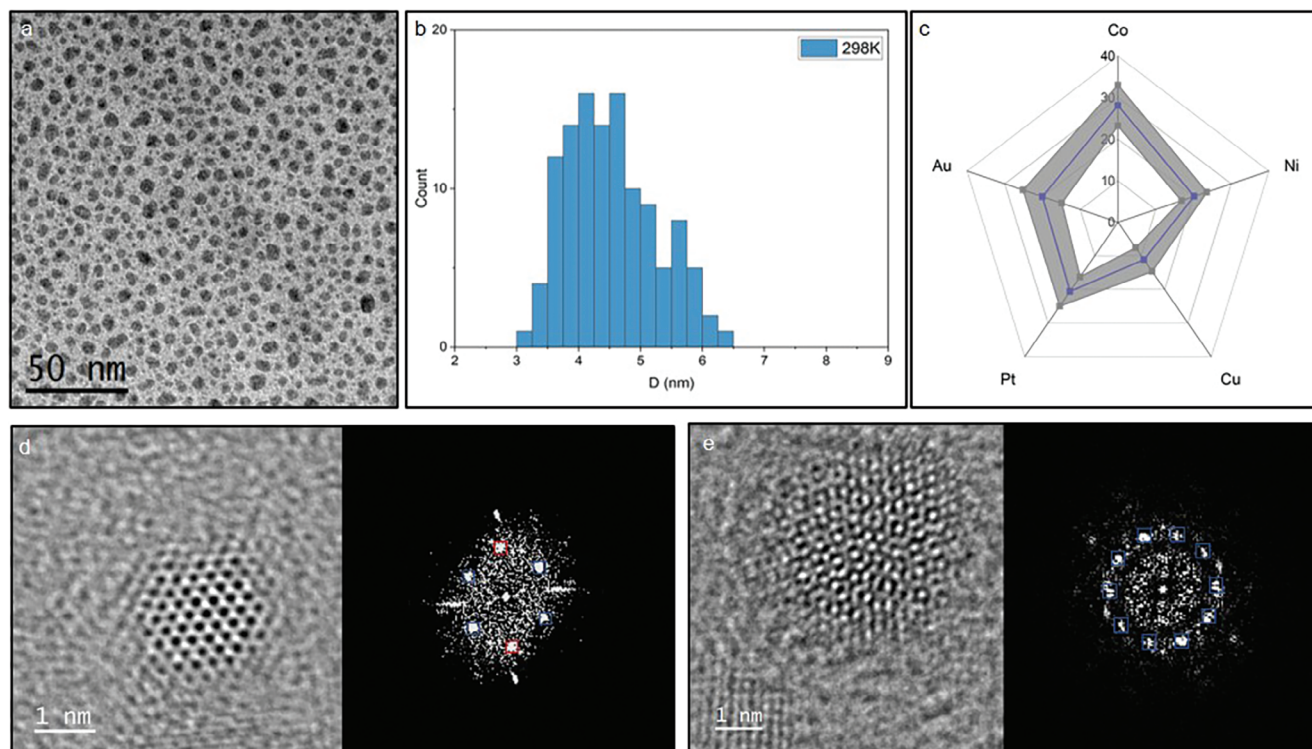


Figure 1. As-prepared CoNiCuPtAu NPs. a–c) HENA fabricated on the amorphous SiN membrane of the heating TEM microchip (a) Bright-field TEM image of NPs. (b) Size distribution of NPs (c) Spider plot of the mean composition of individual NPs measured by STEM-EDX nano-analyses. The mean composition is given in blue and the standard deviation in gray. d,e) HENA fabricated on a thin carbon film. High-resolution TEM images and the corresponding FFT of (d) a monocrystalline HENA observed along its $\langle 110 \rangle$ zone axis and (e) an icosahedral HENA seen along its five-fold axis ($\{111\}$ and $\{200\}$ reflexions of the FCC structure are highlighted in blue and red respectively). The very characteristic contrast of icosahedral FCC NPs observed along their five-fold axis can be compared to the one experimentally observed and simulated in reference.^[8]

and compositional evolution of AuCoCuNiPt NPs from 298 to 973 K. These in situ insights into the thermodynamic properties of AuCoCuNiPt HENA reveal their poor structural stability at high temperature due to atomic diffusion that cannot be considered as sluggish.

2. Results

2.1. In Situ Heating STEM Investigations

AuCoCuNiPt NPs were fabricated by Pulsed Laser Deposition (PLD) directly on the amorphous 50 nm thick SiN film of a heating TEM microchip (Figure 1a, see Experimental Section).^[2a] Particle diameter in the as-prepared sample ranges from 3–6.5 nm with a mean diameter of 4.5 nm (Figure 1b). Prior to heating, the composition of individual NPs was measured by energy dispersive X-ray spectroscopy (EDS) in STEM mode (Figure S1, Supporting Information). The spider plot in Figure 1c shows that particle composition is uniform and $\approx \text{Au}_{20}\text{Co}_{28}\text{Cu}_{11}\text{Ni}_{20}\text{Pt}_{21}$ with a dispersion below 5%. EDS chemical maps revealed the homogenous distribution of the five metals in the NPs (Figure S2a, Supporting Information), and the Face Centered Cubic (FCC) structure of NPs assembly was confirmed by electron diffraction (Figure S2b, Supporting Information). High-resolution imaging was exploited to identify the various shapes of as-grown NPs. To improve data quality, NPs fabricated on 5 nm-thick carbon

film were also analyzed but the shape and structure of HENA are similar on both substrates (Figure S2, Supporting Information). Small monocrystalline FCC NPs with a Wulff shape can be found but most HENA present a multi-twinned structure, including some decahedra and many icosahedra that are clearly identified when they are observed along their two-, three- or five-fold axes. (Figure 1d–e; Figure S2c–j, Supporting Information).

We then combined the high spatial resolution of aberration-corrected STEM and large solid-angle energy dispersive X-ray spectroscopy to follow the size, shape, atomic structure, and composition of these HENA from 298 to 973 K. In Figure 2a, a temperature-series of STEM HAADF images shows the growth and reshaping behavior of an assembly of NP after 15 min at 473, 673, 773, and 973 K. Particle size distribution remains unchanged at 473 K but it is impacted when the temperature reaches 673 K (Figure 2b). Thermo-activated growth mechanisms lead to the progressive loss of the smallest NPs (between 2 and 4 nm) and a widening of the size distribution up to 13 nm, at 973 K. It is also notable that the NP density decreases sharply with increasing temperature. Aside from their growth, NPs also reshape upon annealing. Indeed, the in-plane shape ratio of NPs (i.e., the ratio between the minimum and maximum radii of NPs) increases with temperature, indicating that NPs are evolving from elliptical shapes to spherical ones (Figure 2c).

The growth mechanisms of HENA are here mainly driven by coalescence phenomena that activate when the temperature is

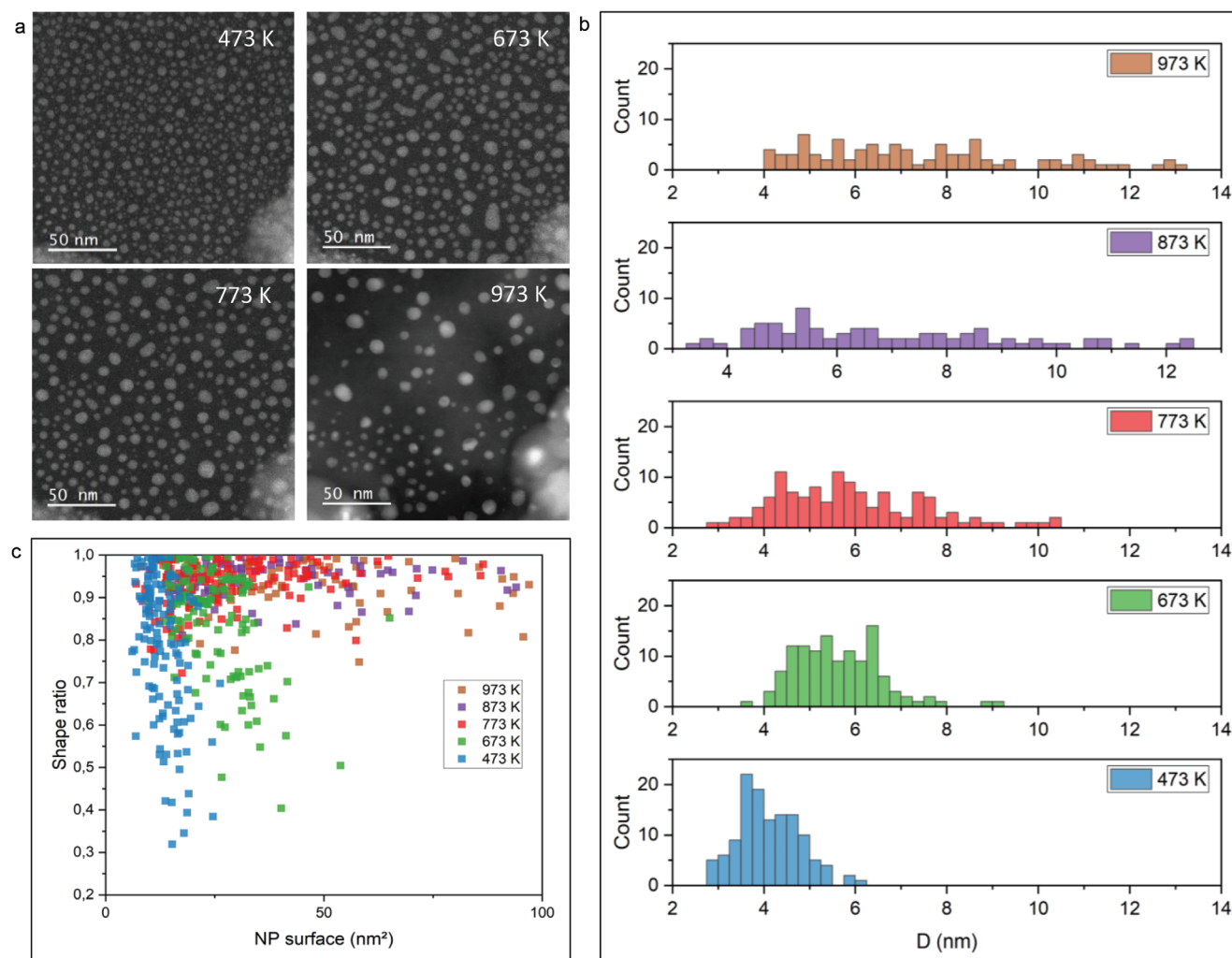


Figure 2. Size and shape evolution during thermal annealing. a) STEM HAADF images showing the evolution of NPs over time at different temperatures, given in the top right corner of each image. All the pictures show the same area 15 min after reaching the temperature step. Evolution of b) the size distribution and c) the shape ratio (i.e., ratio between the in-plane min and max radii) of NPs over temperature.

above 673 K. The size of NPs may also be slightly affected by Ostwald ripening, but as the dissolution of small nanoparticles was never observed during the in situ heating experiments, we assume that the effect of these coarsening mechanisms is negligible. Evidently, coalescence phenomena speed up as the temperature rises. At 773 and 873 K coalescence and reshaping occur in several minutes (Figure 3a,b) while at 973 K it lasts only a few seconds (Figure 3c). Interestingly, phase separation is frequently observed during the coalescence of two sup-5 nm NPs at 773 and 873 K. As illustrated in the STEM HAADF images in Figure 3a,b, high-contrast areas appear at the surface of just merged NPs. These heterogeneous contrasts evidence the segregation of metals with different atomic numbers during coalescence events and they are still observed after the subsequent reshaping of the coalesced nanostructures (Figure S3, Supporting Information). Moreover, STEM-EDS mapping performed at the single NP scale revealed that 5 nm NPs that did not coalesce maintained a well-homogenous distribution of all metals (Figure 4a), but the high-contrast areas observed in larger coalesced NPs are

exclusively made of Au-Cu (Figure 4b). These results suggest that atomic diffusion within the nanostructures at 773 and 873 K tends to segregate Au and Cu at the surface but the resulting Janus configuration can be only observed in coalesced nanostructures in which atom mobility is more important. At 973 K, atomic diffusion is active enough to drive a phase separation in all nanostructures. Indeed, just after raising the temperature from 873 to 973 K, all the NPs reduce in size. As this phenomenon overlaps with the coalescence that is very active at 973 K, this size decrease is not captured in the size distribution measured after 15 min at 973 K (Figure 2b). However as shown in Figure 3d,e, we measure a size reduction of 22% of all the observed NPs within the first minutes at 973 K. Accordingly, a complete loss of gold and copper is evidenced by STEM-EDS, while at lower temperatures the composition of NPs was quite stable (Figure 3f and Figure S4, Supporting Information). The loss of Au and Cu from the NPs at 973 K is most likely due to the segregation of these two metals at the NP surface followed by the melting and evaporation of the Au-Cu shell. It is worth noting that Au and Cu have the lowest

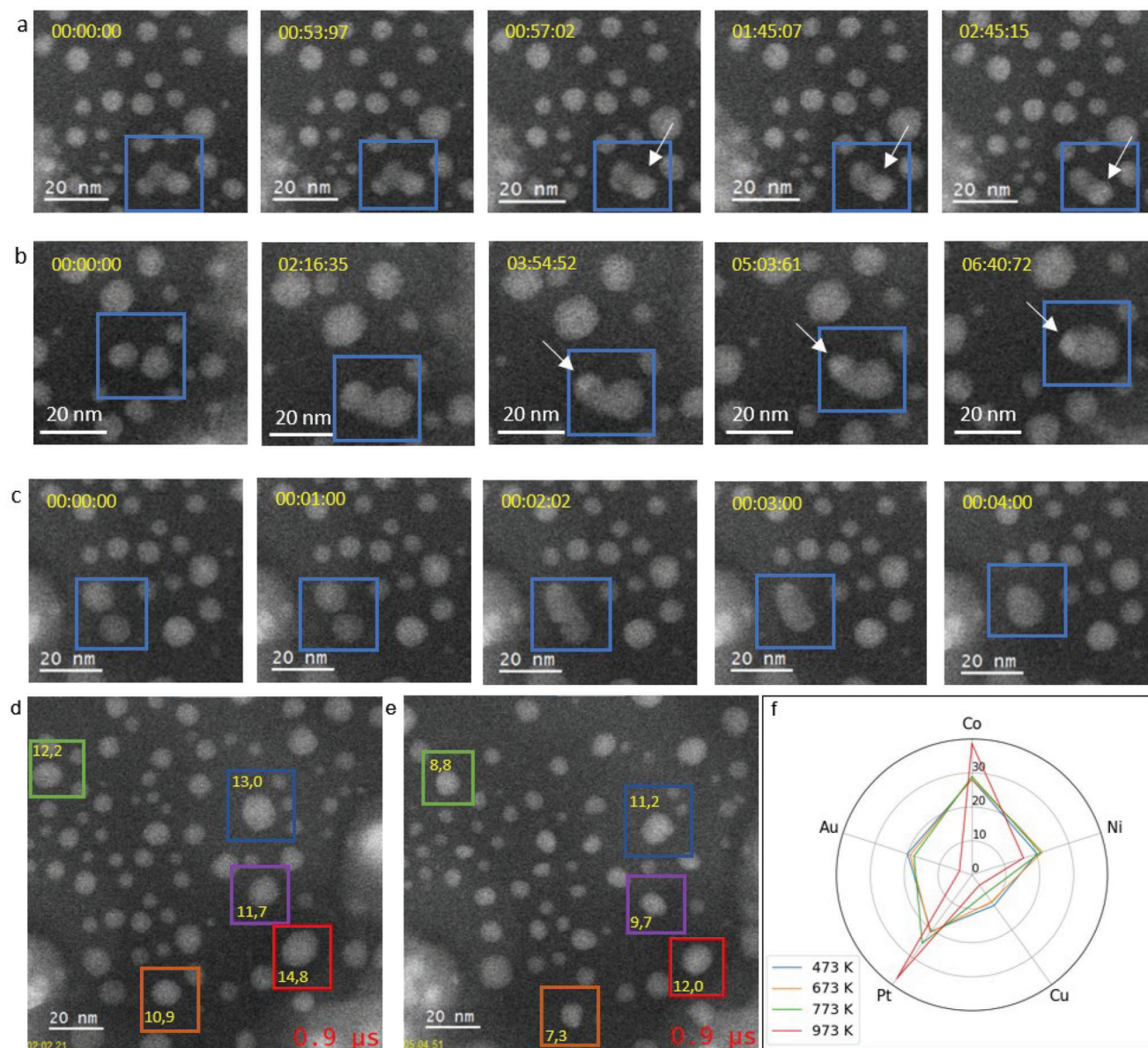


Figure 3. Structural and compositional dynamics of HENA under annealing. Coalescence events of NPs recorded in real-time by STEM HAADF at a) 773 K, b) 873 K, c) 973 K. The acquisition time is indicated in the top left corner of each image. The high Z-contrast areas suggesting the phase separation of metals are indicated by white arrows. STEM HAADF images of the same area recorded d) right after reaching 973 K and e) 3 min later, showing the size decrease of NPs due to the evaporation of gold and copper. The size of some NPs in nm is indicated in both images. f) Evolution of the mean composition of individual NPs measured by STEM-EDX nano-analysis.

melting temperature of the five metals (Table S1, Supporting Information) and the melting of Au, Cu, and AuCu NPs was also observed by TEM at 973 K in previous reports.^[9]

The resulting distribution of Co, Ni, and Pt within the NPs at 973 K is still homogeneous (Figure 4c) and these ternary nanoalloys remain crystalline. Indeed, high-resolution imaging shows that NPs keep their FCC structure from room temperature to 973 K (Figures S5–S7, Supporting Information). Crystal-defect density increases with temperature since the proportion of monocrystalline NPs decreases from 30% at 473 K to 20% at 773 K (Figure S8, Supporting Information). This is likely due to the coalescence phenomena that can lead to the formation of twinned

nanocrystals when two NPs in different orientation merge together.

2.2. Molecular Dynamics Simulations

Our experimental analyses reveal a progressive segregation of gold and copper at the surface with temperature. However, it is not possible to unravel the mechanisms that give rise to such phenomena at the atomic scale. Considering a more simplified system such as binary alloy surfaces or NPs, previous studies have shown that the segregation driving forces are decoupled in terms

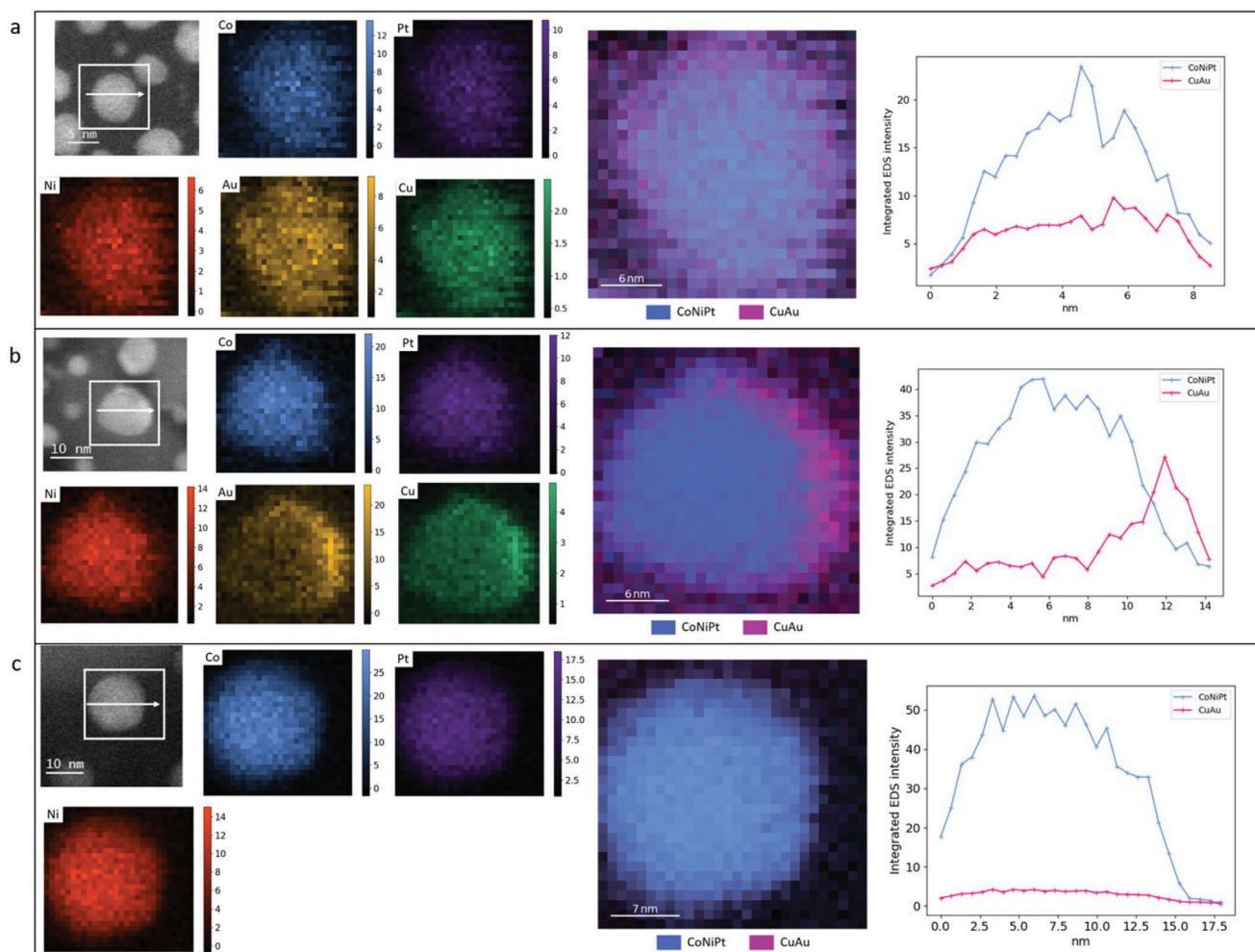


Figure 4. Chemical mapping of HENA at different temperatures. (left to right) STEM HAADF image and EDS elemental maps of the NP framed in white on the STEM HAADF image. EDS elemental maps showing the spatial distribution of CoPtNi and AuCu alloys in the NP and EDS intensity profiles measured along the arrow seen on STEM HAADF images. a) CoNiCuPtAu NP with homogeneous contrast that has not undergone coalescence at 773 K. b) CoNiCuPtAu NP with inhomogeneous contrast that has undergone coalescence at 773 K and c) NP with homogenous contrast at 973 K in which only Co, Ni, Pt are detected.

of three effects, i.e., the cohesion (difference between the surface energies of pure metals), the size (difference between atomic radii) and the alloying (the tendency of the alloy to form homo- or hetero-atomic bounds).^[10] In AuCoCuNiPt HENA, the cohesion effect is clearly favorable to the segregation of gold and copper, which have the lowest surface energies.^[11] Regarding the alloying effect, it is not possible to conclude as easily as the complexity of the system makes it impossible to quantify this contribution. As demonstrated in previous works,^[12] atomic-scale simulations are ideal for tackling this issue. To go beyond and get an insight at the atomic level, we perform Molecular Dynamics (MD) simulations to investigate the thermal properties of CuAuCoNiPt HENA at different temperatures ranging from 400 to 1500 K. **Figure 5a** depicts the structural configurations in case of a 4 nm icosahedral NP containing 2057 atoms when surface concentration for all elements as well as total energy have reached equilibrium (Figure S9, Supporting Information). At low temperatures (500 K), a solid solution is clearly identified: the atoms are dis-

tributed in a fully random distribution within a perfectly crystalline NP displaying (111) facets. As the temperature increases, gold- and copper-enriched zones appear at the surface of the NP. The cross-sectional view further reveals that the presence of these elements alters the local surface structure with a loss of faceting where Au and Cu segregate. In contrast, surface areas where this segregation is not observed have preserved their faceted aspect. Between 1000 and 1200 K, the proportion of copper and gold at the surface becomes predominant and the complete transition from solid to liquid state is observed, as confirmed by the pair correlation function analysis presented in Figure S10 (Supporting Information). To quantify these segregation effects, Figure 5b displays the surface concentration for each element at different temperatures. Below 500 K, we notice that the surface exhibits a structure made of all the elements with a slightly lower proportion of platinum. Then, a change of slope is noticed with an increase in Au at the expense of Pt up to ≈ 900 K. Last, ≈ 1000 K the concentration of nickel and cobalt atoms drops drastically

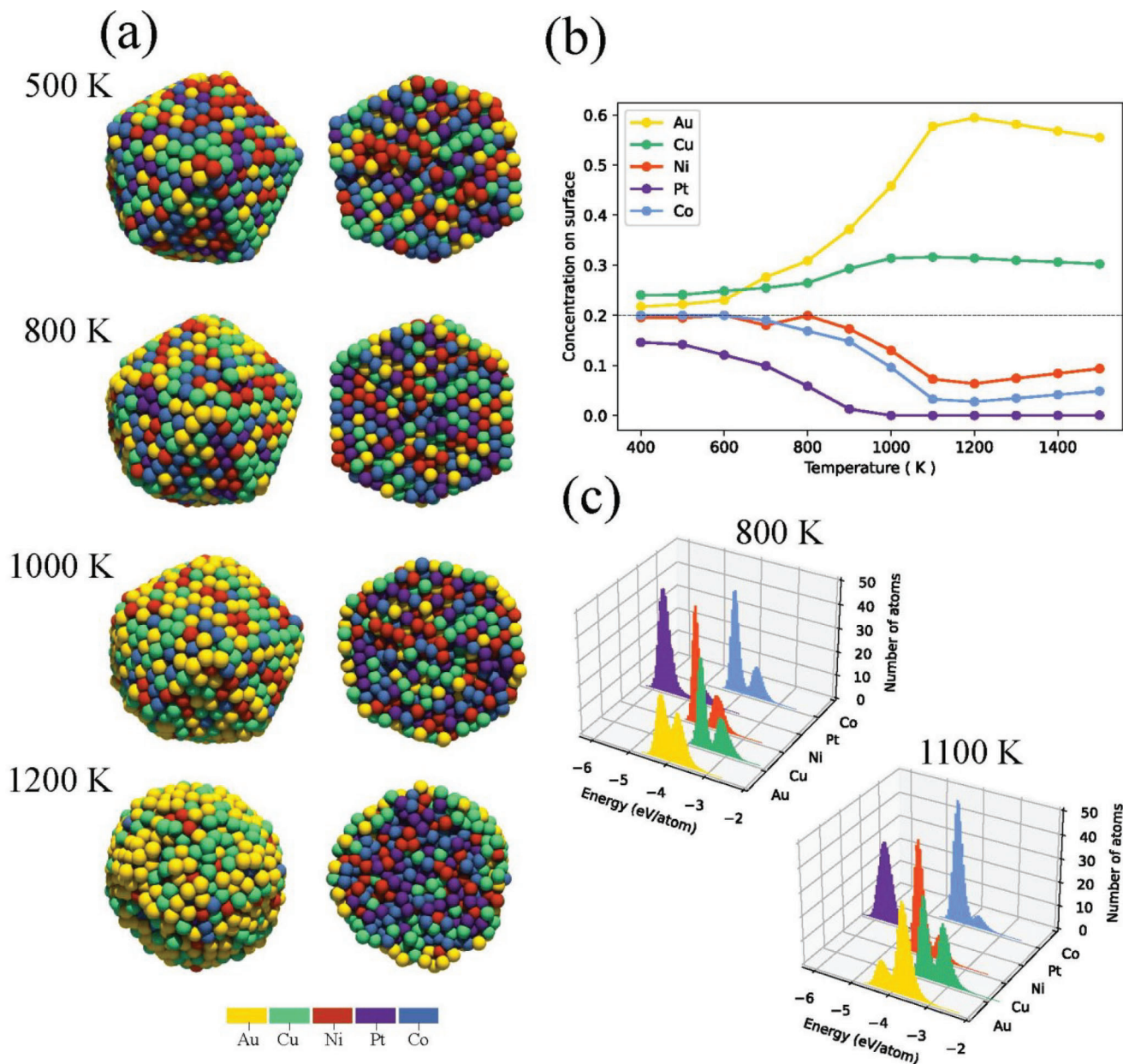


Figure 5. MD simulations investigating the thermal behavior of a 4 nm CoNiCuPtAu NP containing 2057 atoms in an icosahedral structure. a) Equilibrium configurations at different temperatures. The left column presents the shell views and the right one the cross-section views perpendicular to the five-fold axis of the icosahedra. b) Evolution of the surface concentration as a function of the temperature. The dotted line represents the equiatomic concentration. c) Local energies of each metal atoms in histogram forms at 800 and 1100 K.

(below 0.1%), while that of platinum is close to zero giving rise to a surface layer consisting essentially of gold and copper. As seen in Figure S11 (Supporting Information), where the atomic density profiles of Au along the radius of the NP are shown, the enriched zone is not confined to the first layer but extends over the sub-surface layer.

The local atomic energy distributions of all five elements at 800 and 1100 K are presented in Figure 5c. At 800 K, a bimodal distribution consisting of two sub-populations is observed for all metals. The low-energy population corresponds to bulk atoms which have 12 neighbors and the high-energy population cor-

responds to surface atoms which are less stabilized because of their lower coordination. The distribution appears to be similar for all metals, with a larger population of atoms in the core than at the surface of the NP. At 1100 K, nickel and cobalt atoms exhibit almost the same bimodal distribution. It is not surprising that these two elements evolve in the same way since they have very similar structural properties in terms of cohesive energy, lattice parameter, surface energy, and elastic constants.^[13] As for Pt, only a single population can be identified, namely that of the bulk at very low energies. Regarding Au and Cu atoms, the distribution of surface atoms increased to the detriment of the bulk one,

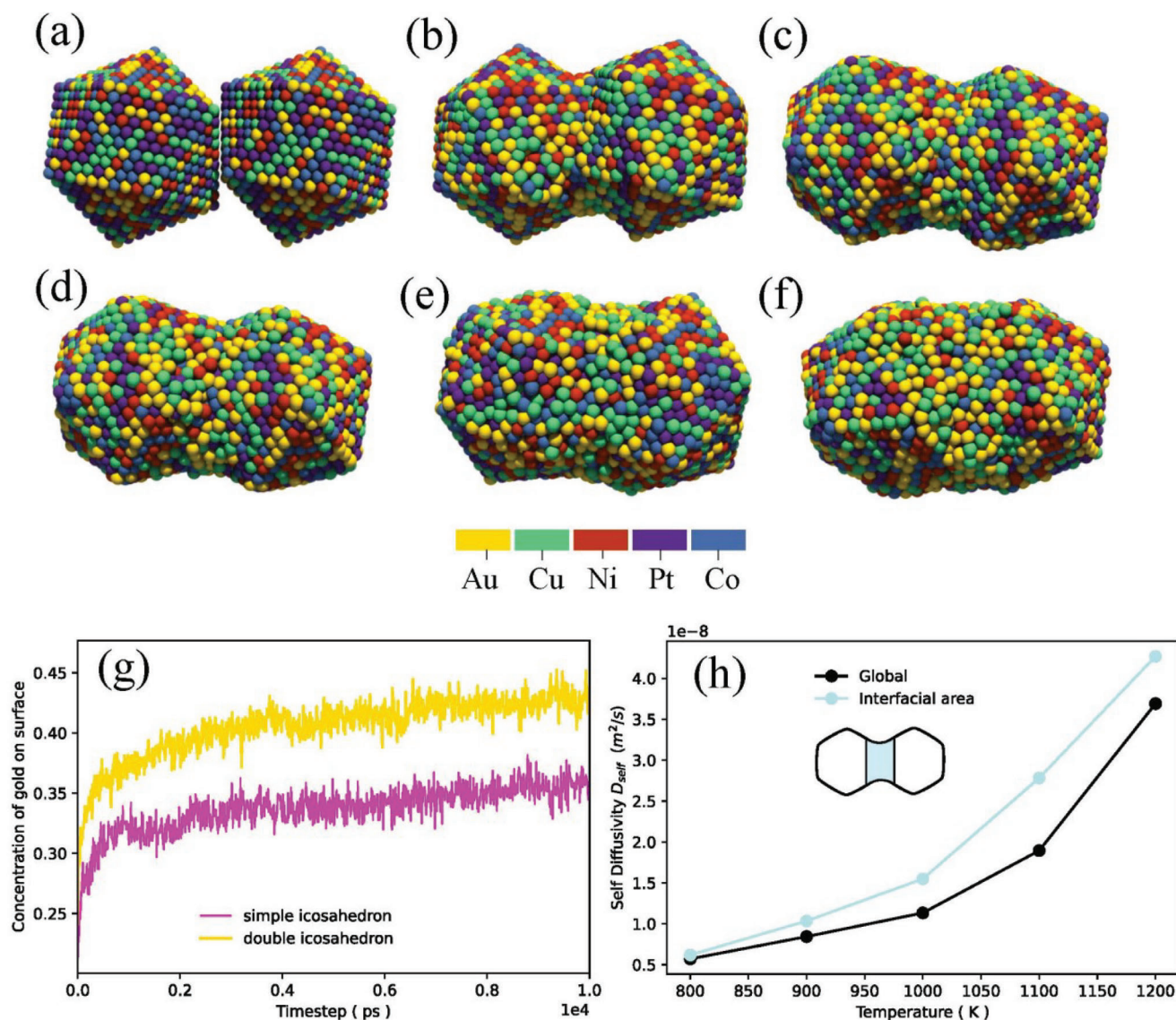


Figure 6. MD simulations investigating the effects of coalescence on the thermal behavior of CoNiCuPtAu NPs. a–f) Snapshots of the atomic configurations during the coalescence of two NPs containing 2057 atoms at 1200 K. g) Comparison of the Au concentration on the surface of a single and a double (i.e. coalesced) icosahedral NP. h) Self-diffusivity of Au atoms as a function of the temperature measured in the whole coalesced NP (global) and in the neck area around the interface between the two coalesced icosahedra as shown in blue in the scheme.

with an almost complete depletion of gold atoms in the NP core. Interestingly, these histograms show that Co, Ni, and Pt atoms in the bulk state have a much lower energy than their counterparts at the surface, thereby favoring strongly their stabilization within the core of the NP. In case of Au and Cu atoms, this difference in energy is much smaller boosting their presence at the surface. In other words, the surface energy, which is directly linked to this difference in energy between surface and bulk atoms, can be considered a major driving force behind segregation. Furthermore, these potential local energy distribution analyses are consistent with in situ TEM observations. Indeed, they show that the surface of AuCoCuNiPt HENA becomes populated by Au and Cu atoms with high local energies as temperature rises, leading very likely to their evaporation at high temperatures. Finally, this thermal

behavior does not seem to be influenced by NP size since similar trends have been highlighted for NPs with diameters ranging from 2.1 to 5.2 nm (Figure S12, Supporting Information).

From experiments, NP coalescence seems to reinforce these segregation effects. To study these mechanisms, we perform MD simulations from two icosahedral NPs close to each other at $T = 1200$ K, each consisting of 2869 atoms with an equiatomic composition. At this stage, it is important to state that constant-temperature simulations (as done here) mimic conditions in which the coalescing clusters dissipate the heat excess due to condensation.^[14] As seen in Figure 6a–f, coalescence occurs, after a few ps, with the formation of a NP whose surface is again highly enriched in copper and gold atoms. Interestingly, the Au segregation is more important in the case of the structure

resulting from coalescence than for the single icosahedron. This is clearly visible in Figure 6g, where the gold concentration at the surface is presented in both situations, showing a more pronounced segregation in the structure after coalescence. Note that only the diffusion of individual atoms is considered here whereas coalescence can involve collective motions of atoms as highlighted in reference.^[14] However, the main conclusions of the present study about neck-enhanced segregation mechanisms are not expected to be altered. To better decipher the segregation mechanisms, we investigate the impact of coalescence on the diffusion of gold atoms within the nanostructures. To this end, the self-diffusivity (D_{self}) of Au was calculated using the Einstein relation (Note S1, Supporting Information) in different areas of the coalesced nanostructure. As the temperature rises, a linear increase in the self-diffusivity of Au atoms is observed until a change in the slope corresponds to the phase transition from solid to liquid (Figure 6h). Furthermore, a higher D_{self} is revealed in the neck region close to the interface between the two NPs. This neck region is known to present a strong interdiffusion activity in order to increase the radius of the neck and reduce its surface curvature.^[15] Therefore, the stronger segregation of Au and Cu observed both experimentally and theoretically in coalesced HENA is most likely due to the reshaping of the nanostructures that boost atomic diffusion.

3. Discussion and Conclusion

Our consistent experimental and theoretical results on the thermal behavior of AuCoCuNiPt prompt us to reconsider the expected thermal stability of HENA at high temperature and more generally their structural properties. At first, according to the Gibbs free energy formula ($\Delta G_{\text{mix}} = \Delta H_{\text{mix}} - T\Delta S_{\text{mix}}$ with T the temperature, ΔH_{mix} and ΔS_{mix} the enthalpy and entropy of mixing, respectively), the increase of the temperature should reinforce the stability of the solid solution phase within the nanostructure. However, both in situ TEM observations and MD simulations reveal that the segregation of Au and Cu at the surface of NPs progressively appears with temperature which clearly indicates that kinetic effects play a key role in the final structure of NPs. Indeed, our study highlights the link between the phase separation and the atomic diffusion within the NPs that obviously speeds up with temperature but also when the nanostructures undergo severe shape and surface restructuring due to coalescence. This result also explains previous reports that revealed the presence of sup-10 nm core-shell nanostructures in the FCC AuCoCuNiPt NP assembly with a mean size below 5 nm fabricated by PLD at 873 K, since coalescence is necessary to reach NP size above 10 nm.^[2a] Thus, our results raise the question: what is the true thermodynamic equilibrium of nanoparticles? Contrary to what their name suggests, the FCC structure observed in the present HENA fabricated by PLD at 873 K is governed by kinetic effects rather than entropy stabilization, because the fabrication time (≈ 12 min) is shorter than the time required to form core-shell structure given the atomic diffusion at 873 K. Under annealing, this out-of-equilibrium FCC solid solution tends toward a phase-separated thermodynamic equilibrium that is stabilized by surface effects since Cu and Au are the metals with the lowest surface energies and their segregation at the surface tends to minimize the total energy. Therefore, this work suggests that if

one intends to explain within a thermodynamic framework the atomic structure of small HENA, a surface energy term must be considered in the Gibbs free energy formula.

Furthermore, it is worth comparing the atomic mobility in AuCoCuNiPt NPs with their metallic and bimetallic sub-nanosystems. Different methods have been used to quantitatively study atomic diffusion in HEA the diffusion, such as interdiffusion studies, tracer studies, or theoretical considerations.^[16] However, the reshaping of NPs and their coalescence are also relevant markers to qualitatively evaluate the impact of surface and bulk mobilities of metal atoms on the NP assembly. Surprisingly, our in situ TEM observations show that reshaping and coalescence processes are observed at only 673 K in the present HENA, whereas they are observed above 873 K in CoPt, AuCu, and Au NPs prepared by PLD on the amorphous substrate.^[9,17] Of course, NP mobility can vary with the nature of the substrate, the size, shape, and crystal structure of NPs, but this direct comparison with metallic and bimetallic NPs fabricated in the same PLD chamber confirms that the lattice distortions and the many crystal defects in the AuCoCuNiPt nanoalloys do not induce sluggish diffusion kinetics. As in bulk alloys,^[16] atomic diffusion in HENA cannot a priori be considered sluggish without experimental or theoretical investigations and this peculiar property very likely depends NP composition, size, and local environment (substrate, matrix, organic ligands...), and the measurement method. From a practical point of view, this study calls into question the use of AuCoCuNiPt HENA for high-temperature applications and more generally reveals the necessity to investigate the thermal behavior of HENA to determine if and how surface effects govern their structural properties, to evaluate their stability and adapt their potential applications.

4. Experimental Section

NP Synthesis: Pulsed laser deposition has been widely used for the synthesis of bimetallic nanoalloys.^[17a,18] The fabrication method permits an alternating deposition of pure metals on the substrate to form nanoalloys of controlled size and composition, using a COMPEX KrF excimer laser beam in a high vacuum chamber. The experimental conditions for the fabrication of sub-10 nm AuCoCuNiPt NPs near the equiatomic composition were recently reported.^[2a] In the present study, the same HENA was directly synthesized on the SiN membrane of a heating MEMS chip commercialized by Protochips for in situ TEM investigations. A substrate temperature of 873 K, a laser frequency of 5 Hz, and a total nominal thickness of 4 nm measured prior to metal deposition were measured with a quartz crystal microbalance. Same HENA were also prepared on an ultra thin amorphous carbon film to acquire high-quality TEM data of as-grown NPs.

In Situ TEM Experiment: In situ heating TEM investigations were performed using the atmosphere sample holder commercialized by Protochips for gas-cell TEM experiments. A single heating chip was used and the microfluidic system was plugged in order to ensure the vacuum sealing of the sample holder while performing TEM investigations with only one SiN membrane along the optical axis. A few tens of nm-large TiO_2 NPs were deposited on the SiN membrane by drop casting prior to the synthesis of HENA in order to facilitate area recognition during in situ TEM experiments. These TEM studies focused on the HENA supported on SiN and were performed with a double aberration-corrected JEOL ARM 200F microscope. Atomic-scale STEM imaging were combined with single NP chemical analysis performed by STEM-EDS. STEM images were acquired with a probe size of 0.08 nm (8C) and a camera length of 8 cm, while STEM-EDX spectra and mapping were acquired with a probe size

of 0.1 nm (5C) with increased beam current for optimal X-ray generation and detection. Temperature was controlled with a ceramic heating device that supports the SiN membrane.^[19] TEM observations were performed at 473, 673, 773, 873, and 973 K and a heating rate of 1 K s⁻¹ was used to reach these temperature steps. As HENA continuously evolves at high temperatures, the analysis time at each temperature was limited to 45 min. We had to go back to 298 K after each temperature step to perform EDS analyses to avoid the adverse effect on the X-ray detection efficiency of IR radiation emitted from the heating device at high temperatures.^[20] Wiener and ABSF filters were applied on high resolution STEM Images to improve the visibility of nanocrystal structure.

Computational Methods: Molecular Dynamic (MD) simulations are performed using the open source Large-scale Atomic/Molecular Massively Parallel Simulators (LAMMPS) package.^[21] In order to investigate Cu-Au-Co-Ni-Pt NPs, the Embedded Atom Method (EAM) potential derived by Zhou et al. is used to describe the interaction between different atomic pairs.^[22] To validate this approach, we carried out comparisons with another EAM-type potential, i.e., Foiles et al.^[23] More details are given in the (Note S2, Supporting Information). We choose to consider an equiatomic system and not the concentration observed experimentally, in order to be close to the high entropy alloy definition. Note that this choice does not alter the important conclusions of this study (Figure S13, Supporting Information). Furthermore, we perform MD simulations based on the Verlet algorithm with a timestep of 1 fs combined with the Nose-Hoover thermostat at a given temperature (including a temperature damping parameter equal to 100 timesteps) for the integration of equations of motion in the NVT canonical ensemble. The size of the supercell in all directions was large enough to avoid artifacts due to interaction between images caused by periodic boundary conditions. Prior to dynamic simulations, the NPs were relaxed at 0.01K using a conjugated gradient method until convergence of the total energy to get the equilibrium configuration. Regarding the MD simulations at a given temperature, the evolution of both the total energy and the surface concentration of each element is monitored and considered complete when these two quantities reach a plateau (Figure S9, Supporting Information). Finally, all the quantities analyzed (surface concentration, atomic energy, diffusion, etc.) are derived from the averages of a set of configurations (corresponding to approximately the last 50 ns of simulations) where the plateau is reached. Knowing that thermodynamic equilibrium is difficult to achieve at low temperature, complementary simulations were reproduced at lower heating rate (1K ns⁻¹). Results are presented in Figures S14 and S15 (Supporting Information). Furthermore, different initial configurations were considered to ensure that the initial preparation did not affect our results (Figure S16, Supporting Information).

As seen experimentally, HENA prepared by PLD are FCC multi-twinned nanostructures and many NPs present an icosahedral shape. As defects can influence atomic diffusion and metal segregation,^[24] we investigate NPs in an icosahedral structure having a five-fold symmetry and containing 309, 923, 2057, and 3871 randomly distributed on the FCC multi-twinned lattice. This corresponds to cluster sizes ≈ 2 to 5 nm close to the range explored experimentally. From a theoretical point of view, we made the choice to study icosahedral NPs that are similar to those observed experimentally but beyond size effects,^[25] the shape of NPs can also have an influence on their thermal stability.^[26] This last point is illustrated in Figure S17 (Supporting Information) where the gold segregation in case of an icosahedron and truncated octahedron of the same size is presented.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

atomistic simulations, high entropy alloy nanoparticles, in situ TEM, phase transition, sluggish diffusion, thermodynamics

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